5. Exploration of matrix multiplication

Bookmark this page

Calculator

Recitation due Feb 14, 2024 20:30 IST

Exploration

2/2 points (graded)

Let $\mathbf{A}$ and $\mathbf{B}$ both be $\mathbf{3} \times \mathbf{3}$ matrices. So in particular, we can take the matrix product $\mathbf{AB}$ and the matrix product $\mathbf{BA}$.

Let $\mathbf{A}$ be the matrix representing the linear transformation that rotates all vectors in $\mathbb{R}^3$ 90 degrees clockwise about the positive y-axis.

$$A = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Let $\mathbf{B}$ be the matrix representing the linear transformation that projects all vectors in $\mathbb{R}^3$ onto the xy-plane.

$$B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

What is $\mathbf{AB} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

(Enter as a column vector. Separate entries using a comma: for example type **[a , b , c]** for the column vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$.)

[0, 0, 0]

✔ Answer: [0, 0, 0]

What is $BA \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

(Enter as a column vector. Separate entries using a comma: for example type **[a , b , c]** for the column vector $\begin{pmatrix} a \\ b \\ c \end{pmatrix}$.)

[-1, 0, 0]

✔ Answer: [-1, 0, 0]

Submit You have used 1 of 5 attempts

Answers are displayed within the problem

MATLAB activity

Generate two square, random matrices $\mathbf{A}$ and $\mathbf{B}$ of the same size using MATLAB online.

Calculator

To do this, type `rand(10)` for example to generate a 10x10 square matrix whose entries are random numbers between 0 and 1. You may wish to shift this by 1/2 and multiply by any number $2n$ to get random entries between $-n$ and n: e.g. `(rand(10)-0.5)*200`.

Find $\mathbf{AB}$.

Find $\mathbf{BA}$.

Does $\mathbf{AB} = \mathbf{BA}$?

Conjecture based on exploration

1/1 point (graded)

In general, for two $n \times n$ matrices $\mathbf{A}$ and $\mathbf{B}$, does $\mathbf{AB} = \mathbf{BA}$?

- ○ Yes.
- ● No.

✔

Solution:

Matrix multiplication is not commutative! That is $\mathbf{AB} \neq \mathbf{BA}$ for almost all matrices. This somewhat surprising fact makes the study of matrices even more interesting. This is often the first example of a non-commutative multiplication people encounter.

Submit You have used 1 of 1 attempt

Answers are displayed within the problem

5. Exploration of matrix multiplication

Hide Discussion

Topic: Unit 2: Linear Algebra, Part 2 / 5. Exploration of matrix multiplication

Add a Post

Show all posts by recent activity

There are no posts in this topic yet.

✖

Previous Next

Calculator

Calculator